

Intelligent Conversational AI Chatbot

Automated Customer Support for Billing Disputes & Offer Recommendations

NLP · Deep-Learning NLU · Hub-and-Spoke Agents · Reinforcement Learning

The Problem

- Telecom support is dominated by two repetitive intents: **“why is my bill wrong?”** and **“is there a better deal?”**
- Human agents are costly, inconsistent, and hold times hurt satisfaction.
- These flows are rule-heavy and latency-sensitive — ideal for AI automation.

Goal: an AI agent that understands free-text queries, acts on the real account, resolves the issue, and escalates hard cases — without fabricating any charge or offer.

Objectives

1. Automated, real-time conversational support using NLP.
2. Improve response quality & efficiency (↑ satisfaction, ↓ response time).
3. Deep-learning NLU + NLG for context-aware, human-like interactions.
4. Reinforcement learning for continuous improvement from feedback.
5. Evaluate: accuracy, intent precision, resolution rate, latency, satisfaction.

What the Chatbot Does



Understand your bill

Explains changes line-by-line.



Dispute a charge

Shows evidence; resolves via credit, waiver or ticket.



Get offers

Personalized, RL-optimized recommendations.

- **Identity validation** before any account data is shown.
- **Hub-and-spoke** handoff from front-line agent to retention specialist.
- **Grounded NLG** — optional local LLM only rephrases, never invents facts.

Architecture — “Thick Brain, Thin UI”

Per-turn pipeline: **NLU** → **Policy** → **Hub routing** → **Spoke flow** → **Recommendation (+RL)** → **NLG**

Layer	Does
Presentation (Streamlit)	Auth, chat, live signals, KPIs
Brain (Coordinator + spokes)	Understand, route, decide, act
Data & config (JSON, .env)	Accounts, offers, models, provider

Two AI spokes — **Aria** (billing & offers) and **Ray** (retention) — never talk directly; only the hub moves control.

Hub-and-Spoke in Action

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 **Marcus Rivera**

BRONZE · VALUE · 6mo ·

ESSENTIAL_5G_35

Current bill (2026-04)

\$62

↑ \$27 vs last month

🔔 Live signals

Intent

HAG...

↑ 100%

Sentiment

😐

↑ NEU

Active spoke: Retention (Ray)

Handoff reason: haggle

Resolution: in_progress

🧠 Config

LLM phrasing (template (no LLM))

> 🎯 RL offer learning (M5)

> 📊 KPIs (M6 monitoring)

Deploy

⋮

intent BILL_EXPLAIN · sentiment 😊 · policy default · 13ms

👍

👎

🗣️ i never used that much data

👤 Aria · Billing & offers

I understand. I re-checked and metering shows 3.1 GB used in the window 2026-03-25 to 2026-04-05, beyond your plan cap. Here's what I can do: I can apply a one-time 50% goodwill waiver on the overage (\$10 off), or open a formal dispute ticket. Which would you prefer?

intent FACTUAL_DISPUTE · sentiment 😊 · policy factual_dispute · 1ms

👍

👎

🗣️ that is too expensive can you do better

👤 Ray · Retention specialist

I hear you want a better deal, and I have more room than the front-line does. I can offer you 'One-time 50% waiver on current bill overage', worth \$10 — one-time goodwill waiver on the current overage charge. This is a genuine retention rate. Want it?

⚡ Handed off to Retention specialist (Ray) — reason: haggle

intent HAGGLE · sentiment 😊 · policy default · 2ms

👍

👎

Ask about your bill, a charge, or offers...

➤

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NLU — Deep-Learning Understanding (M3)

- **Intent** — 14 classes, TF-IDF → **MLP neural network**, keyword fallback.
- **Sentiment** — POS/NEU/NEG MLP → signed score + intensity.
- **Entities** — regex slot-filling: charge ids, amounts, dispute topic, yes/no.
- Fused into a single **TurnFrame** consumed by the coordinator.

Every turn shows live intent, sentiment, and policy signals in the UI sidebar.

NLG — Grounded & Pluggable (M4)

- Flows emit a deterministic **template + grounded facts**.
- Optional **local LLM (Ollama · llama3.2)** rephrases in the persona voice.
- Provider set in `.env`: `ollama` `template` `openai` `anthropic`
- LLM is told **not to invent** numbers → no hallucinated bills/offers.
- Any LLM failure → automatic template fallback (demo never breaks).

Reinforcement Learning (M5)

- Epsilon-greedy **bandit**: each offer is an arm with shows/accepts.
- Reward: **accept = 1, decline = 0**; value Q = accepts / shows.
- Ranking blends learned value: $\text{score} = 0.7 \cdot \text{base} + 0.3 \cdot Q$.
- Feedback from accept/decline and 👍/👎; **policy persists** across sessions.

Effect: offers customers actually accept float to the top over time — continuous improvement from feedback loops.

Offer Recommendation (UC-02)

<<

Ethan Walker

GOLD · HIGH_VALUE · 47mo · UNLIMITED_PREMIUM_85

Current bill (2026-04)

\$122

↑ \$30 vs last month

Live signals

Intent

PRO...

↑ 100%

Sentiment

😞

↑ NEU

Active spoke: ● Billing (Aria)

Resolution: open

Config

LLM phrasing (template (no LLM))

> 🧠 RL offer learning (M5)

> 📊 KPIs (M6 monitoring)

Deploy ⋮

Support chat

Aria · Billing & offers

Hi Ethan! I'm Aria, your support assistant. I can explain your bill, help with a disputed charge, or find offers for you. What can I help with today?

🤖 do you have any offers for me

Aria · Billing & offers

Based on your account, my top recommendation is 'Add-on 5 GB data free for 2 months' (worth about \$12) — data usage is HIGH — extra data avoids overage. Would you like me to apply it?

Add-on 5 GB data free

\$12

↑ score 0.816

Gold loyalty — 10% off

\$24

↑ score 0.689

intent: PROMO_INQUIRY · sentiment: 😞 · policy: default · 2ms

👍

👎

Ask about your bill, a charge, or offers...

➤

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Results & Evaluation

Metric	Value	Source
Sentiment accuracy (held-out)	~83%	MLP, stratified split
Intent accuracy (held-out)	~62%*	MLP, small seed corpus
Resolution rate	100%	Test scenarios
Avg per-turn latency	~2 ms	Template mode
CSAT proxy	👍 / 👎 feedback	Session logs

*Limited by the small PoC seed corpus; runtime trains on full data + keyword fallback. Reproducible via `python evaluate.py`.

Tech Stack

UI — Streamlit chat

NLU — scikit-learn MLP + TF-IDF

NLG — templates + Ollama LLM

RL — epsilon-greedy bandit

Data — synthetic JSON

Config — .env provider switch

Runs entirely on a laptop — one command: `./run.sh`

Conclusion & Future Scope

Delivered: an end-to-end conversational agent that understands queries, resolves billing disputes, recommends & applies offers, escalates via hub-and-spoke, learns from feedback, and is fully observable — meeting all five objectives.

Future scope:

- Mine NLU training data from real chat logs; add a transformer NLU.
- Wire live CRM / billing systems in place of synthetic adapters.
- Contextual RL policy over customer features; supervisor approval workflow.
- Proactive validation of promised actions to cut repeat contacts.

Project Milestones & Outcomes

Milestone	Outcome
M1 Data Collection	Synthetic dataset for 3 customers exercising every use case
M2 Data Preparation	Labelled 14-intent + 3-class sentiment corpus + preprocessing
M3 NLU Model	Trained MLP intent + sentiment + entities (~83% sentiment)
M4 NLG Module	Grounded templates + local Ollama LLM, hallucination-free
M5 Reinforcement Learning	Epsilon-greedy bandit, persisted, improves offer ranking
M6 Deployment & Monitoring	One-command Streamlit app, live signals + KPIs

Future State — Roadmap

Stronger AI

Transformer-based NLU; corpus mined from real chat logs; retrieval-augmented answers for open FAQs.

Live integration

Replace synthetic adapters with production CRM, billing & ticketing connectors.

Smarter learning

Contextual RL policy over customer features; real supervisor-approval workflow.

Scale & reach

Cloud deployment behind an autoscaling endpoint; responsive mobile experience; voice channel.

Future Use Cases

1 · Proactive offers on login (SOP-driven)

Evaluate the customer's profile against Standard-Operating-Procedure rules at login and surface the single most relevant eligible offer unprompted — reactive → proactive relationship manager.

2 · Post-application validation agents

After a credit / waiver / offer is applied, a background agent re-reads the authoritative state to confirm fulfilment and auto-remediates discrepancies — closes the promise-to-fulfilment loop.

3 · Call reduction via proactive network-telemetry fixes

Ingest throttling / outage / coverage signals and pre-emptively credit or re-plan affected customers before they contact support — shifts the operator from reactive to preventive.

Thank You

Demo: `./run.sh` → <http://localhost:8501>

Repository & docs: [README.md](#) · [architecture/](#) · [docs/demo/demo.mp4](#)